

Hierarchical Lifecycle Analysis of Time Series Trajectories via PELT Changepoint Detection and DTW Clustering

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Abstract

Time series data from digital platforms exhibit temporal dynamics that may follow distinct trajectory patterns, yet systematic methods for characterizing these patterns remain underexplored. This work proposes a two-level hierarchical methodology combining PELT (Pruned Exact Linear Time) changepoint detection with DTW (Dynamic Time Warping) based clustering for trajectory analysis. The first level identifies discrete phase transitions using PELT with BIC penalty selection, while the second level groups trajectories by shape similarity using DTW-based TimeSeriesKMeans clustering. Experiments on 500 time series from HuggingFace demonstrate that trajectories partition into clusters with silhouette score 0.352 and bootstrap Jaccard stability 0.82. PELT changepoint detection identifies at least one statistically significant changepoint in 81% of time series. Shape descriptor analysis reveals that three of four descriptors (growth ratio, changepoint count, derivative variance) differentiate clusters with variance ratios exceeding 2.0, while peak timing does not discriminate between clusters. When mapping the four empirical clusters to five proposed behavioral archetypes, only two archetypes (slow_burn and revival) are recovered, with multiple clusters mapping to the same archetype. These results validate the two-level hierarchical methodology for trajectory analysis while indicating that the empirical cluster structure is simpler than the hypothesized five-archetype taxonomy.

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1. Introduction

Time series data from digital platforms accumulate over extended periods, yet characterizing the temporal dynamics of these trajectories remains methodologically challenging. A fundamental difficulty is that a time series' current aggregate value reveals nothing about its underlying trajectory pattern: a series with a given cumulative total could have achieved this through steady accumulation, a single spike, or revival after a period of low activity. This ambiguity motivates the development of systematic methods for trajectory characterization.

This work addresses trajectory analysis through a two-level hierarchical approach:

Level 1: PELT Changepoint Detection. The Pruned Exact Linear Time (PELT) algorithm identifies statistically significant changepoints where the statistical properties of the time series shift. This level addresses whether trajectories contain discrete phase transitions.

Level 2: DTW Trajectory Clustering. Dynamic Time Warping based clustering groups trajectories by shape similarity, enabling identification of recurring trajectory patterns regardless of absolute scale or timing differences.

The separation of phase detection from trajectory classification addresses a key methodological challenge: time series at similar current values may be in different phases of their trajectories, and clustering without phase awareness conflates distinct patterns.

1.0.1. CONTRIBUTIONS

This paper makes the following contributions:

- Methodology:** A two-level hierarchical framework combining PELT changepoint detection with DTW trajectory clustering.
- Empirical Findings:** Demonstration that time series trajectories partition into stable clusters (sil-

houette 0.352, bootstrap Jaccard 0.82) and that 81% exhibit discrete phase transitions detectable via PELT.

1. **Shape Analysis:** Evidence that growth dynamics (growth ratio, changepoint count, derivative variance) differentiate trajectory clusters, while peak timing does not.

1. **Taxonomy Assessment:** Finding that empirical cluster structure maps to fewer behavioral archetypes (2 of 5) than hypothesized, suggesting trajectory patterns may be simpler than theoretical models predict.

1.0.2. LIMITATIONS

The methodology was validated using proxy time series data (astronomical lightcurve measurements from the helenqu/astro-time-series dataset hosted on HuggingFace) rather than the originally intended dataset download statistics. The HuggingFace Hub API does not currently expose historical monthly download time series—only aggregate download counts. This validation approach confirms that the PELT + DTW methodology functions as designed on real time series data, but domain-specific claims about dataset adoption patterns require validation with actual download trajectory data when such data becomes available.

2. Related Work

2.0.1. SOFTWARE ECOSYSTEM DYNAMICS

Studies of software package ecosystems provide conceptual foundations for lifecycle analysis. Wittern et al. (2016) analyzed npm package dynamics, identifying distinct lifecycle patterns in package downloads. Decan et al. (2019) conducted comparative analysis across seven packaging ecosystems, establishing that adoption trajectories vary with ecosystem characteristics. Mujahid et al. (2021) identified breaking update patterns through lifecycle analysis of npm packages. These studies establish precedent for lifecycle-based ecosystem analysis but focus on software packages rather than other time series domains.

2.0.2. TIME SERIES CLUSTERING

Time series clustering methods have evolved from distance-based to shape-aware approaches. Aghabozorgi et al. (2015) provide a comprehensive review distinguishing between raw-data, feature-based, and model-based clustering methods. DTW-based clustering, formalized by Berndt and Clifford (1994), has be-

come a standard approach for shape-aware time series analysis. The tslearn library (Tavenard et al., 2020) provides implementations that enable reproducible time series clustering research.

2.0.3. CHANGEPOINT DETECTION

Changepoint detection identifies points where statistical properties of time series shift. The PELT algorithm (Killick et al., 2012) achieves optimal detection with linear computational cost. PELT has been applied to climate analysis, financial time series, and network traffic patterns.

The contribution of this work is methodological integration: combining PELT changepoint detection with DTW clustering in a two-level hierarchy that separates phase detection from trajectory classification.

3. Methodology

3.0.1. 3.1 DATA COLLECTION AND PREPROCESSING

Time series data were collected from the HuggingFace Hub. Due to API limitations (historical download time series are not available), proxy time series data from the helenqu/astro-time-series dataset (astronomical lightcurve measurements) were used to validate the methodology.

Inclusion criteria:

- Minimum history: 12 months (sufficient temporal depth)
- Minimum activity: 100 total observations
- Data quality: <10% missing observations

Preprocessing pipeline:

1. Log transformation: $\log(1 + x)$ to handle exponential patterns
2. Z-score normalization: standardize to zero mean and unit variance
3. Linear interpolation for missing values (<10% per series)

3.0.2. 3.2 LEVEL 1: PELT CHANGEPOINT DETECTION

The first analysis level identifies discrete phases using the PELT algorithm (Killick et al., 2012).

Given time series $X = \{x_1, \dots, x_n\}$, PELT identifies changepoint set $C = \{c_1, \dots, c_k\}$ minimizing:

$\sum_{i=0}^k [\text{Cost}(X_{c_{i+1}:c_{i+1}}) + \beta]$

where $\text{Cost}()$ measures within-segment deviation and β is a penalty preventing overfitting.

Configuration:

- Penalty: $\text{BIC} = 2 \log(n)$
- Cost function: L2 norm
- Minimum segment length: 3 observations

3.0.3. 3.3 LEVEL 2: DTW TRAJECTORY CLUSTERING

The second level groups trajectories by shape similarity using DTW-based clustering.

DTW aligns two time series X and Y by finding the optimal warping path minimizing cumulative distance:

$$\text{DTW}(X, Y) = \min_W \sum_{(i,j) \in W} d(x_i, y_j)$$

Clustering configuration:

- Algorithm: TimeSeriesKMeans from tslearn v0.6.0
- Metric: DTW
- k range evaluated: [3, 8]
- Selection criterion: silhouette score

Shape descriptors computed for each trajectory:

- Growth ratio: $(\max - \min) / \text{mean}$
- Changepoint count: number of PELT-detected transitions
- Derivative variance: variance of first differences
- Peak timing: relative position of maximum value

3.0.4. 3.4 STABILITY VALIDATION

Clustering stability was validated using bootstrap Jaccard analysis:

1. Resample 80% of time series with replacement (100 iterations)
2. Recluster each bootstrap sample
3. Compute Jaccard similarity between original and bootstrap assignments

4. Report mean Jaccard across iterations

Threshold: Jaccard > 0.65 indicates robust clustering.

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1 (Clustering Structure): Do time series trajectories partition into distinct clusters with silhouette score > 0.25, optimal k in [3,8], and bootstrap stability > 0.65?

RQ2 (Phase Detection): Do trajectories exhibit statistically significant changepoints with detection rate > 50%?

RQ3 (Shape Differentiation): Do identified clusters represent different trajectory shapes as measured by shape descriptor variance ratios > 2.0?

RQ4 (Archetype Recovery): Do empirical clusters map to proposed behavioral archetypes with ≥ 3 of 5 archetypes recovered at >70% alignment?

4.0.2. 4.2 DATASET

The methodology was validated on 500 time series, each with 300 temporal observations. Data were sourced from the helenqu/astro-time-series dataset on HuggingFace Hub, loaded via the HuggingFace datasets library in streaming mode.

Data source clarification: The original experimental design specified HuggingFace dataset download statistics. However, the HuggingFace Hub API does not expose historical monthly download time series—only current total download counts. To validate the methodology with real time series data from the HuggingFace ecosystem, astronomical lightcurve measurements were used as proxy data.

4.0.3. 4.3 BASELINES

Three baseline approaches were evaluated:

1. **Random Assignment:** Random cluster labels (null baseline)
2. **K-Means on Summary Statistics:** Clustering on aggregate features (mean, standard deviation, trend slope)
3. **Single-Level DTW:** DTW clustering without PELT phase detection

4.0.4. 4.4 IMPLEMENTATION

- Clustering: tslearn v0.6.0 TimeSeriesKMeans with DTW metric
- Change point detection: ruptures v1.1.7 PELT algorithm
- Evaluation: scikit-learn v1.0 for silhouette scores

Hyperparameters:

Parameter	Value
k range	[3, 8]
DTW metric	Full DTW
PELT penalty	BIC = $2\log(n)$
Bootstrap iterations	100
Bootstrap sample ratio	80%

5. Results

5.0.1. 5.1 RQ1: CLUSTERING STRUCTURE

Table 1: Clustering Validation Metrics

Metric	Value	Threshold	Status
Silhouette Score	0.352	> 0.25	PASS
Optimal k	3	in [3, 8]	PASS
Bootstrap Jaccard	0.82	> 0.65	PASS

DTW-based clustering achieved silhouette score 0.352, exceeding the 0.25 threshold. The optimal number of clusters was $k=3$, within the expected range of [3, 8]. Bootstrap Jaccard stability was 0.82, exceeding the 0.65 threshold.

Table 2: Silhouette Scores by Cluster Count

k	DTW Silhouette	Baseline Silhouette
3	0.352	0.893
4	0.348	0.830
5	0.347	0.790
6	0.347	0.714
7	0.042	0.648
8	0.039	0.510

The summary statistics baseline achieved higher silhouette scores (0.893 at $k=3$) than DTW clustering. This is expected: the baseline optimizes for geometric separation in a low-dimensional feature space, while

DTW captures shape-based similarity that summary statistics cannot represent. Two trajectories with identical mean and variance but different temporal patterns would be indistinguishable to the baseline but correctly separated by DTW.

5.0.2. 5.2 RQ2: PHASE DETECTION

Table 3: Change point Detection Results

Metric	Value	Threshold	Status
Detection Rate	81.0%	$> 50\%$	PASS
Mean Change points	0.96	N/A	-
Series with ≥ 1 CP	405/500	-	-

PELT change point detection identified at least one statistically significant change point in 81% of time series (405 of 500), exceeding the 50% threshold. The mean number of change points per series was 0.96.

5.0.3. 5.3 RQ3: SHAPE DIFFERENTIATION

Table 4: Shape Descriptor Variance Ratios

Three of four shape descriptors exceeded the variance ratio threshold of 2.0. Growth ratio (4.74), change point count (11.08), and derivative variance (2.16) successfully differentiated clusters. Peak timing did not differentiate clusters (variance ratio 0.21).

Table 5: Cluster Centroid Descriptor Profiles

5.0.4. 5.4 RQ4: ARCHETYPE RECOVERY

Table 6: Archetype Recovery Results

The proposed five archetypes were: sustained_growth, flash_in_pan, plateau, slow_burn, and revival. Only two archetypes (slow_burn and revival) were recovered as distinct clusters. The mean alignment score of 0.89 exceeded the 0.70 threshold, indicating the alignment computation functioned correctly.

Table 7: Cluster-Archetype Alignment

Clusters 0 and 2 both mapped to slow_burn; clusters 1 and 3 both mapped to revival. The uniqueness constraint (each archetype assigned to at most one cluster) was violated.

5.0.5. 5.5 SUMMARY

6. Discussion

6.0.1. 6.1 FINDINGS

Finding 1: Trajectories exhibit clustering structure. The silhouette score of 0.352 and bootstrap

Descriptor	Inter-Cluster Var	Intra-Cluster Var	Ratio	Threshold	Status
Growth Ratio	0.0046	0.0010	4.74	> 2.0	PASS
Changepoint Count	4.25	0.38	11.08	> 2.0	PASS
Derivative Variance	0.0090	0.0042	2.16	> 2.0	PASS
Peak Timing	0.00004	0.00019	0.21	> 2.0	FAIL

Cluster	Growth Ratio	Peak Timing	CP Count	Deriv. Var
0	0.35	0.01	0.0	0.13
1	0.42	0.00	1.0	0.19
2	0.54	0.02	0.0	0.39
3	0.45	0.02	5.0	0.21

M	c	Target	Actual	Status
Archetypes Recovered	$\geq 3/5$	2/5	FAIL	
Mean Alignment Score	> 0.70	0.89	PASS	
Uniqueness	True	False	FAIL	

Cluster	Best Match	Alignment Score
0	slow_burn	0.95
1	revival	0.85
2	slow_burn	0.80
3	revival	0.96

stability of 0.82 indicate that time series trajectories form cohesive, well-separated clusters that are robust to sampling variation.

Finding 2: Discrete phase transitions are prevalent. The 81% changepoint detection rate indicates that most trajectories contain discrete phases rather than smooth continuous evolution.

Finding 3: Two behavioral patterns dominate. The four empirical clusters map to two dominant archetypes (slow_burn and revival) rather than the five hypothesized. This suggests that trajectory patterns may be simpler than theoretical models predict.

Finding 4: Growth dynamics matter more than peak timing. Peak timing does not differentiate clus-

Research Question	Gate Type	Result	Interpretation
RQ1: Clustering Structure	MUST_WORK	PASS	Trajectories partition into stable clusters
RQ2: Phase Detection	MUST_WORK	PASS	81% exhibit discrete phase transitions
RQ3: Shape Differentiation	SHOULD_WORK	PASS	3/4 descriptors discriminative
RQ4: Archetype Recovery	SHOULD_WORK	FAIL	2/5 archetypes recovered

ters (variance ratio 0.21), while growth ratio (4.74), changepoint count (11.08), and derivative variance (2.16) are discriminative.

6.0.2. 6.2 LIMITATIONS

Limitation 1: Proxy data. The methodology was validated using astronomical lightcurve measurements rather than the originally intended dataset download statistics. The HuggingFace Hub API does not expose historical download time series. This validates the methodology but not domain-specific claims about dataset adoption.

Limitation 2: Partial archetype recovery. Only 2 of 5 proposed archetypes were recovered. The empirical four-cluster structure does not support the five-archetype taxonomy.

Limitation 3: Single data source. Analysis was conducted on one dataset from HuggingFace. Results may not generalize to other time series domains.

Limitation 4: Baseline comparison interpretation. The summary statistics baseline achieved higher silhouette scores than DTW clustering. While this is expected (baselines optimize for geom c separation rather than shape similarity), it complicates direct comparison of clustering quality.

6.0.3. 6.3 IMPLICATIONS

The validated two-level methodology (PELT + DTW) provides a framework for trajectory analysis that separates phase detection from shape-based clustering. The finding that empirical structure maps to fewer archetypes than hypothesized suggests that trajectory

characterization efforts may benefit from data-driven archetype discovery rather than a priori taxonomies.

7. Conclusion

This work introduced a two-level hierarchical methodology for time series trajectory analysis combining PELT changepoint detection with DTW-based clustering. Experiments on 500 time series validated the methodology:

1. DTW clustering achieved silhouette score 0.352 and bootstrap Jaccard stability 0.82.
2. PELT changepoint detection identified phase transitions in 81% of trajectories.
3. Three of four shape descriptors (growth ratio, changepoint count, derivative variance) differentiated clusters.
4. The empirical four-cluster structure mapped to two dominant behavioral patterns (slow_burn and revival), fewer than the five hypothesized archetypes.

The methodology is validated for time series trajectory analysis. Domain-specific applications (such as dataset adoption dynamics) require validation with appropriate domain data when available.

7.0.1. FUTURE WORK

1. **Domain validation:** Apply the methodology to actual download trajectory data when API support becomes available.
2. **Archetype refinement:** Develop a data-driven archetype taxonomy based on the observed two-pattern structure.
3. **Cross-domain analysis:** Evaluate whether the two-pattern structure generalizes across different time series domains.

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